Supporting information for *The Selling closure: complete reindexing of measured lattices*

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Table and section numbers of the main text are cited as there; equations of the main text are cited by number. The marks [z*nn*] name the check files of Section S4.

S1. Catalogues of shapes and specialisations

Cell shapes. Table S1 lists the shapes of the ten sorts with more than one, each as the cell with the most right angles and no acute angle. The three shapes of a primitive monoclinic lattice are the three cell choices of *International Tables* (Hahn, 2006, Section 2.2.16); the extra shapes of primitive orthorhombic, tetragonal and cubic lattices are monoclinic-looking cells with two right angles; C-centred orthorhombic has two primitive shapes; the boundary sorts of body-centred orthorhombic, C-centred monoclinic and triclinic have two or three cells with a single right angle each, for triclinic the known instability of the reduced cell when an angle is near 90° (Andrews *et al.*, 1980).[z49]

Table S1. *The cell shapes of the sorts with more than one, for the cells named; each shape is given as the cell with the most right angles and all angles* $\geq 90°$*, shortest edges first when several qualify and b unique for the monoclinic-looking ones, with the number n of members (up to swaps) that have it. Centred cells are given by their primitive cells; the boundary sorts are specified by their conorms in the* S^6 *slot order of Table 4 of the main text, in* $Å^2$*.*[z49]

sort	lattice	n	shape
cP	30 Å	4	30, 30, 30; 90, 90, 90
		12	30, 30, 42.4; 90, 135, 90
tP	30, 30, 50 Å	4	30, 30, 50; 90, 90, 90
		4	30, 30, 58.3; 90, 121.0, 90
		8	30, 50, 42.4; 90, 135, 90
oP	30, 40, 55 Å	4	30, 40, 55; 90, 90, 90
		4	30, 40, 62.6; 90, 118.6, 90
		4	30, 55, 50; 90, 126.9, 90
		4	40, 55, 50; 90, 143.1, 90
oS	C 30, 40, 50 Å	2	25, 50, 25; 90, 106.3, 90
		4	25, 50, 30; 90, 126.9, 90
mP	30, 40, 50 Å, 100°	2	30, 40, 50; 90, 100, 90
		2	30, 40, 53.7; 90, 113.4, 90
		2	50, 40, 53.7; 90, 146.6, 90
oI (O3)	900, 1600, 0, 900, 1600, 0	1	50, 50, 50; 111.1, 129.8, 90
		2	42.4, 50, 50; 90, 115.1, 115.1
mC (M2B)	900, 1600, 0, 900, 1600, 2500	1	50, 50, 70.7; 104.7, 116.9, 90
		1	50, 50, 65.6; 105.9, 105.9, 90
mC (M3)	900, 1600, 0, 2500, 1600, 0	1	50, 50, 64.0; 120, 90, 111.1
		2	50, 56.6, 58.3; 90, 108.0, 124.4
aP (A2)	900, 1600, 0, 2500, 400, 1200	1	36.1, 60.8, 64.0; 114.3, 90, 114.2
		1	36.1, 56.6, 64.0; 116.2, 90, 101.3
aP (A3)	900, 1600, 0, 2500, 400, 0	1	36.1, 50, 53.9; 90, 101.9, 119.9
		1	44.7, 50, 53.9; 90, 99.6, 135.7
		1	36.1, 44.7, 58.3; 90, 115.3, 104.4

Specialisations. The conorms of any member, in any labelling, are integer combinations of the reduced cell's conorms, hence linear in the free parameters of the sort (the letters of Table 4 of the main text); imposing another sort's zeros and equalities on one member is a homogeneous linear system in those parameters, and a solution with non-negative parameters, a non-singular lattice and a resulting lattice of the target sort and no higher is a specialisation. Table S2 lists the conditions of codimension one, on 36 pairs of sorts.[z50] Each is a parameter vanishing or two parameters becoming equal. The classical values appear: body-centred tetragonal is face-centred cubic at $c/a = \sqrt{2}$ and body-centred cubic at $c/a = 1$; C-centred orthorhombic is hexagonal at $b = \sqrt{3}\,a$ and tetragonal at $b = a$; rhombohedral is cubic at $\alpha = 60°$, $90°$ and

109.47°. The monoclinic conditions come in threes, one per shape: with unique axis c and plane lattice a, b, the lattice is C-centred orthorhombic when $|a| = |b|$, $|a| = |a+b|$ or $|b| = |a+b|$, and primitive orthorhombic when $a \perp b$, $a \perp (a+b)$ or $b \perp (a+b)$ — the textbook condition $a = c$ read in each of the three cell choices. The table also shows which sorts are one condition from a much higher symmetry: C-centred monoclinic from tetragonal, body-centred tetragonal and rhombohedral; the body-centred orthorhombic boundary sort from face-centred cubic; triclinic on a wall from C-centred monoclinic.

Table S2. *All specialisations of codimension one between the 24 sorts. Conditions are on the letters of the sort's pattern in Table 4 of the main text; "shapes" is the number of distinct cell shapes of the starting sort through which the conditions are reached, and the last column reads the condition in the cell of the conventional description where that is short.*[z50]

from	to	conditions	shapes	reading
aP A1	aP A2	any letter $= 0$	1	a right angle appears
aP A2	aP A3	$v = 0$	1	a second right angle
aP A2	mC M2A	$s = t$, $r = u$	1	
aP A2	mP M4	r, s, t or $u = 0$	1	
aP A3	mC M3	any two letters equal	3	one per shape
mP M4	oP O5	$r = 0$, $s = 0$, $t = 0$	3	$a \perp b$, $a \perp (a+b)$, $b \perp (a+b)$
mP M4	oS O4	$s = t$, $r = t$, $r = s$	3	$\lvert a\rvert = \lvert b\rvert$, $\lvert a\rvert = \lvert a+b\rvert$, $\lvert b\rvert = \lvert a+b\rvert$
mC M1A	oF O1A	$r = t$	1	
mC M1A	oI O2	$u = 0$	1	
mC M1A	oS O4	$r = 0$, $t = 0$	1	
mC M1B	mC M2B	$t = 0$, $u = 0$	1	boundary sort
mC M1B	mC M3	$r = 0$, $s = 0$	1	boundary sort
mC M1B	oF O1A	$r = s$	1	
mC M1B	oI O1B	$t = u$	1	
mC M2A	oI O2	$r = t$	1	
mC M2B	oI O3	$t = 0$	1	
mC M2B	tI T2	$r = s$	1	one step to tI
mC M2B	tP T5	$r = 0$, $s = 0$	1	one step to tP
mC M3	oI O3	$r = t$	1	
mC M3	hR R3	$r = s$, $s = t$	1	one step to hR
oP O5	tP T5	$r = s$, $r = t$, $s = t$	1	$a = b$, $a = c$, $b = c$
oS O4	tP T5	$r = 0$	1	$b = a$
oS O4	hP H4	$r = s$	1	$b = \sqrt{3}\,a$
oI O1B	oI O3	r, s or $t = 0$	1	boundary sort
oI O1B	tI T1	$r = s$, $r = t$, $s = t$	1	two edges equal
oI O2	tI T2	$r = s$	1	
oI O3	cF C3	$r = s$	1	one step to cF
oF O1A	tI T1	$s = t$	1	
oF O1A	tI T2	$s = 0$, $t = 0$	1	
tP T5	cP C5	$r = s$	1	$c = a$
tI T1	cI C1	$r = s$	1	$c/a = 1$
tI T1	cF C3	$s = 0$	1	$c/a = \sqrt{2}$ from below
tI T2	cF C3	$s = 0$	1	$c/a = \sqrt{2}$ from above
hR R1	cP C5	$r = 0$	1	$\alpha = 90^\circ$
hR R1	cI C1	$r = s$	1	$\alpha = 109.47^\circ$
hR R3	cF C3	$r = s$	1	$\alpha = 60^\circ$

Table S3. *Four walks on the measured metric, and the construction of Appendix B of the main text, against brute force on 80 cells with zero conorms at awkward positions (orthorhombic, cubic, plate, monoclinic and hexagonal zero patterns), with a band τ on the conorms at* 0.1 *and* 0.5 *of* $\lambda_1/32$.[z45]

procedure	cells with a wrong set
flip wherever $\lvert p\rvert \le \tau$, keep only cells within the band	13
flip wherever $\lvert p\rvert \le \tau$, filter at the end	13
flip wherever $\lvert p\rvert \le 2\tau$, keep only cells within the band	13
flip wherever $\lvert p\rvert \le 2\tau$, filter at the end	2
idealise at 4τ, enumerate the exact closure, filter	0

S2. Walks on the measured metric, and the benchmark

The repository directory `manuscript/benchmark` holds 780 measured cells with exact references and true operators, in five families: 75 cells measured in a different shape of the same lattice; 660 lattices placed 0.5 to 8 times the error from a condition of Table S2; 20 cells whose only reduced superbase is a difference-wall member; 16 needles and plates with angular errors; 9 cells of the boundary sorts. Each case also carries its Niggli cell. `evaluate.py` scores the conorm-band methods of Appendix B of the main text and `evaluate_strain.py` the strain-norm procedure, by whether the true operator is proposed from one reduction of the measured cell; the results are in `results.txt` and `results_strain.txt`.

S3. The operators alone

The closure can be described without any lattice. An operator $\mathbf{W}$, written as the new basis in the coordinates of the reduced cell, acts on the six conorms by a fixed integer 6×6 matrix $\mathbf{A_W}$, since every conorm of $\mathbf{W}F_0$ is an integer combination of the conorms of F_0. The reduced cells are the points of the orthant $\{p \geq 0\}$ in conorm space; Selling (1874) showed that every lattice has a conorm vector in this orthant, and it is unique up to the 48 swaps except on the faces (Delaunay, 1933; Conway & Sloane, 1992), and the closure of a lattice is the set of points of its orbit that lie in the orthant: one in the interior, several on the faces. So $\mathbf{W}$ is an operator of the closure of every lattice with zero pattern Z exactly when $\mathbf{A_W}$ maps the face $\{p \geq 0,\ p_Z = 0\}$ into the orthant, which is a sign condition on the entries of $\mathbf{A_W}$ and nothing else. Enumerating integer matrices under that condition gives the operator set of each face without a lattice: $48 \times$ 1, 2, 3, 6 and 16 matrices for no zero, one zero, two zeros on disjoint edges, two zeros on a path and three zeros, and their union, over all placements of the faces, is a set $\mathcal{U}$ of $6096 = 48 \times 127$ integer matrices with entries in $\{-3, \ldots, 3\}$. $\mathcal{U}$ is

closed under inversion, every element has a labelling whose basis vectors have entries in $\{-2,\ldots,2\}$, and $\mathcal{U}$ contains every operator between two reduced cells of one lattice, whatever the lattice: an operator out of any member is an operator out of a reduced cell of that member's zero pattern, Lemma A1 of the main text confines the members of a closure to superbase coordinates of modulus at most 7 relative to that cell, and the exhaustive enumeration of that window finds only members with coordinates in $\{-1,0,1\}$, so every such operator has entries in $\{-1,0,1\}$ in the labelling of its source and lies within the window enumerated.[z52,z56] The three-zero face enumerated is the triangle; its path placement gives the same operators relabelled. Up to relabelling at both ends $\mathcal{U}$ has nine kinds of element, of flip length 0 to 4 (Table S4).

Table S4. *The nine kinds of operator between Selling-reduced cells, up to relabelling at both ends; "matrices" counts the integer matrices of the kind in $\mathcal{U}$, "flips" its length as a flip word, and the last column the zero patterns in whose closures it occurs.*[z52]

kind	matrices	flips	occurs with
1	48	0	every pattern (the swaps)
2	288	1	every pattern with a zero
3	1152	2	two zeros on a path; three zeros
4	1152	3	three zeros
5	1152	3	three zeros
6	576	3	two zeros on a path; three zeros
7	576	4	three zeros
8	576	4	three zeros
9	576	4	three zeros

Why four. For a lattice with $a \perp b \perp c$ the sixteen members are four orthogonal superbases, differing by the sign of one axis, and twelve with a face diagonal. A flip from an orthogonal cell always gives a cell with a face diagonal, so changing the sign of one axis takes three flips, out, across and in, and the diagonal cells attached only to the far orthogonal cell take one more: from (a,b,c) the members lie at flip distances $0,1,2,3,4$ in numbers $1,3,6,3,3$, and four is the diameter of the largest face.[z46]

S4. Computational checks

The mark [lib] refers to `agentsg/tests/test_strain_closure.py`; each mark [z*nn*] refers to `manuscript/appendix_z/test_znn_*.py` in the `agentsg` repository.

mark	what is checked
z36	conorm shift identity; constants by zero pattern and frame (Appendix B)
z42	the S^6 distance (CS6Dist) on the worked pair, where it agrees with the closure
z44	the constants 4 and 8; class gap and excess; Theorem B5 against brute force; the difference-wall configuration
z45	reduction with a widened boundary from random starts; the walks of Table S3; the worked cell at $\tau = 20$; the conorms of Table 6 of the main text
z46	the monoclinic example; closures of the 24 sorts; shapes = orbits; twofolds generate; difference-wall members; idealising at 2τ; nesting
z47	reindexing under a declared error with the conorm band: membership, true operator, guarantee (Appendix B)
z49	the cell shapes of Table S1
z50	the codimension-one specialisations between the 36 pairs of sorts and the shapes that carry them (Table S2)
z51	the seven BLDConograph sample cells and the program's distances (Table 5 of the main text)
z52	the operator sets of the faces from the matrices alone; the universal set $\mathcal{U}$, its inversion closure and window; the nine kinds (Table S4); generation by flips
z53	the 24 sorts as the closed sets of equalities of Proposition A5, with their symmetry orders
z54	every column of Table 4 of the main text computed from the operators alone
z55	the conorm-band construction at bands from $\lambda_1/32$ to $\lambda_1/6$
z56	Lemma A1: the obtuse superbases of every zero pattern inside the window
z57	bands per slot: the needle on which idealisation in the frame of the reduced cell fails
z58	the strain norm: projection, enumeration and filter on the needle, two plates and 240 random cells against brute force; distances to the sorts and their monotonicity along Table S2
z59	Lemma A6 and Theorem A7: the ball bound, the adjacency radii of Table 7 of the main text, the layers of 127 and 2755, the filtered first layer below δ_1, the plates between δ_1 and δ_2
z60	the strain depths of Table 6 of the main text and the growth of the widened set; the sort spectra of Fig. 2 of the main text; Table 5 of the main text; the prices of the second indexing classes in Fig. 4 of the main text and the β scan; centred references in conventional settings; the $\epsilon/(1-\epsilon)$ bound of Corollary A8; δ_1 at the cap for the spectrum cells; the single projection against the union on the needle
lib	the library: layers and sizes, the adjacency radius, reindexing with and without the chirality filter, the scale fit, the indexing classes of Table 8 of the main text with the anomalous and chiral counts, the price spectrum of the worked cell

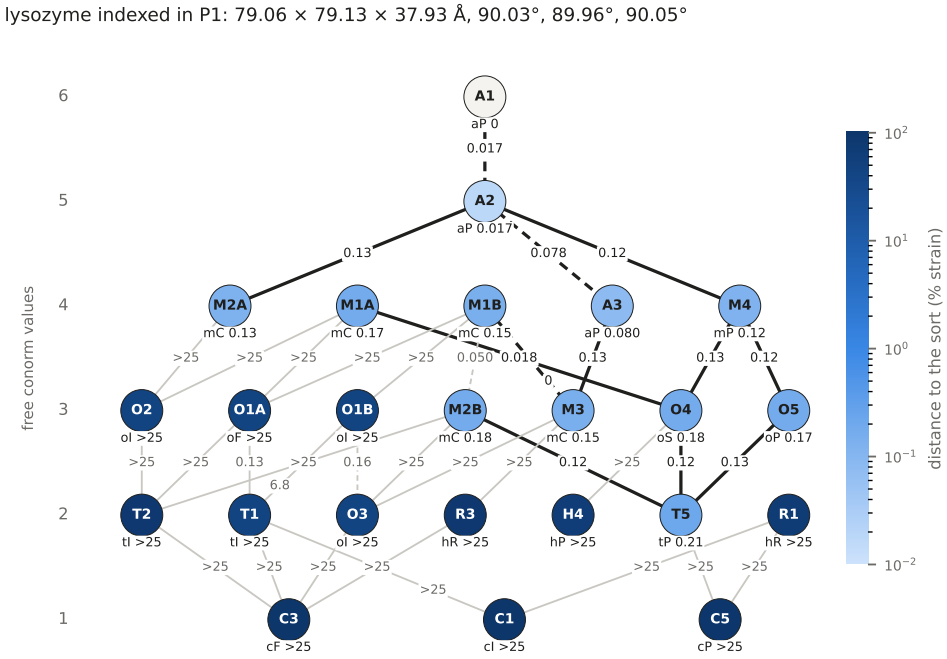


Fig. S1. The 24 Delone sorts as a graph for a tetragonal lysozyme cell (79.1 × 79.1 × 37.9 Å) indexed in $P1$ with an added error, drawn as Fig. 2 of the main text. The twelve sorts from aP to tP lie within 0.22%; every other sort is more than 25% away.[z60]

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